

Homework 3

Submission 1, Spring 2026

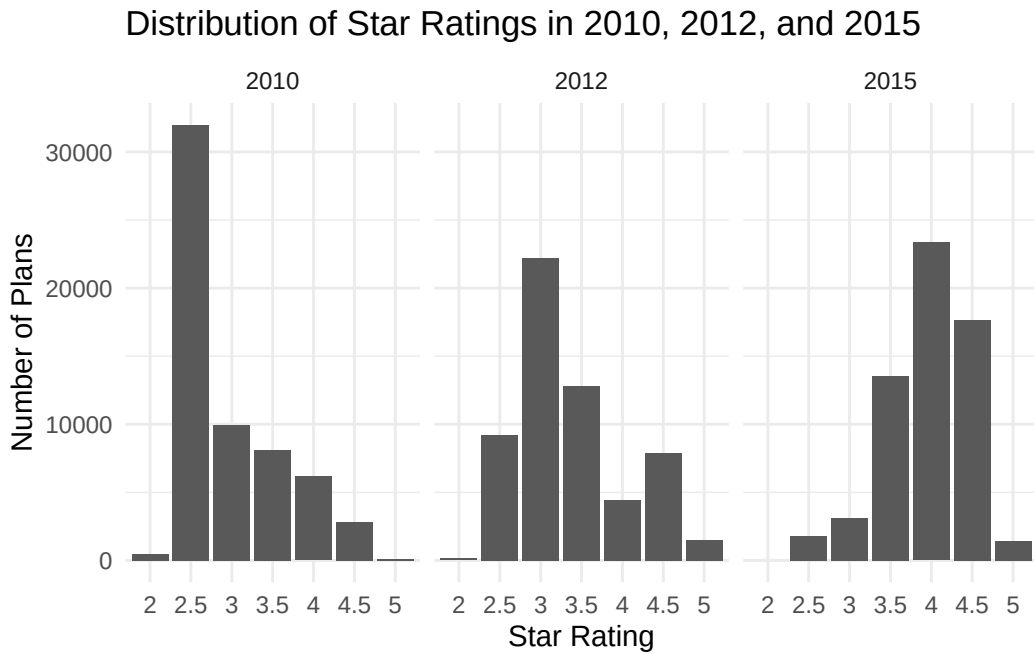
Sophie Kim

```
# A tibble: 6 x 5
  year `Mean Star Rating` `Mean Enrollments` `Mean Market Share` `Total Plans`
  <dbl>         <dbl>         <dbl>         <dbl>         <int>
1  2010             2.97             256.             0.0658        108222
2  2011             3.26             343.             0.0855         68003
3  2012             3.36             377.             0.0856         67246
4  2013             3.55             396.             0.0815         67835
5  2014             3.81             437.             0.08          62344
6  2015             3.96             470.             0.0787         65499
```

Problem 2

```
# A tibble: 6 x 4
  year `Mean Enrollments` `Mean Market Share` `Total Plans`
  <dbl>         <dbl>         <dbl>         <int>
1  2010             107.             0.0673         48643
2  2011             168.             0.09          12754
3  2012             173.             0.0567          9077
4  2013             214.             0.06           3661
5  2014             258.             0.0557          3708
6  2015             314.             0.0405          4680
```

Problem 3



Problem 4

	2010	2011	2012	2013	2014	2015
Intercept (2.5 stars)	0.0634	0.0813	0.0876	0.0488	0.0606	0.0764
3 Stars	0.0126	0.0009	0.0202	0.0653	0.0450	0.0104
3.5 Stars	0.0005	0.0348	0.0072	0.0322	0.0221	0.0167
4 Stars	-0.0036	-0.0116	-0.0250	0.0264	0.0219	-0.0048
4.5+ Stars	-0.0076	-0.0221	-0.0196	0.0105	0.0026	0.0071

Problem 5

filter first

```
# A tibble: 3 x 2
  `Star Rating`    `Number of Plans`
  <chr>            <int>
1 Rounded up to 3      2217
2 Rounded up to 3.5    1462
3 Rounded up to 4       698
```

build raw rating then filter

```
# A tibble: 5 x 2
  `Star Rating`      `Corresponding Number Of Plans`
  <chr>                <int>
1 Rounded up to 3      9918
2 Rounded up to 3.5    8091
3 Rounded up to 4      4284
4 Rounded up to 4.5     771
5 Rounded up to 5       30
```

Problem 6

Sharp RD estimates using local polynomial regression.

Number of Obs.	26038	
BW type	Manual	
Kernel	Uniform	
VCE method	HCO	
Number of Obs.	12800	13238
Eff. Number of Obs.	5094	858
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	0.125	0.125
BW bias (b)	0.125	0.125
rho (h/b)	1.000	1.000
Unique Obs.	12800	13238

	Point	Robust Inference		
	Estimate	z	P> z	[95% C.I.]
RD Effect	0.046	-0.802	0.422	[-0.033 , 0.014]

Sharp RD estimates using local polynomial regression.

Number of Obs.	26038
BW type	Manual
Kernel	Uniform

VCE method	HCO	
Number of Obs.	18122	7916
Eff. Number of Obs.	1164	1003
Order est. (p)	1	1
Order bias (q)	2	2
BW est. (h)	0.125	0.125
BW bias (b)	0.125	0.125
rho (h/b)	1.000	1.000
Unique Obs.	18122	7916

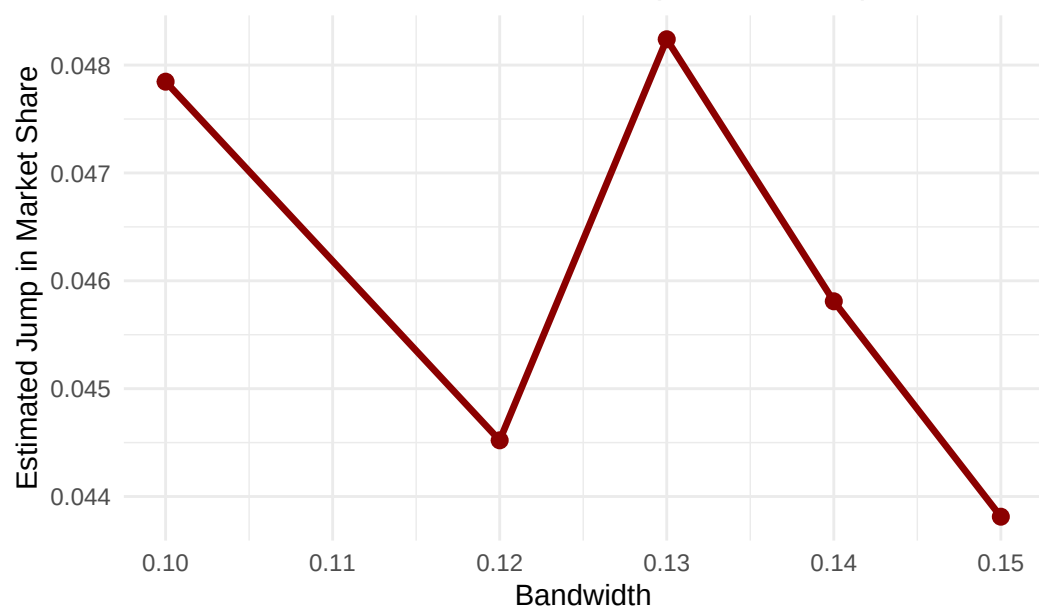
```
=====
              Point      Robust Inference
              Estimate      z      P>|z|      [ 95% C.I. ]
-----
RD Effect    -0.049    -5.202    0.000    [-0.089 , -0.040]
=====
```

```
# A tibble: 2 x 2
  Cutoff      Estimate
  <chr>      <dbl>
1 3 Stars vs. 2.5 Stars  0.0461
2 3.5 Stars vs. 3 Stars -0.049
```

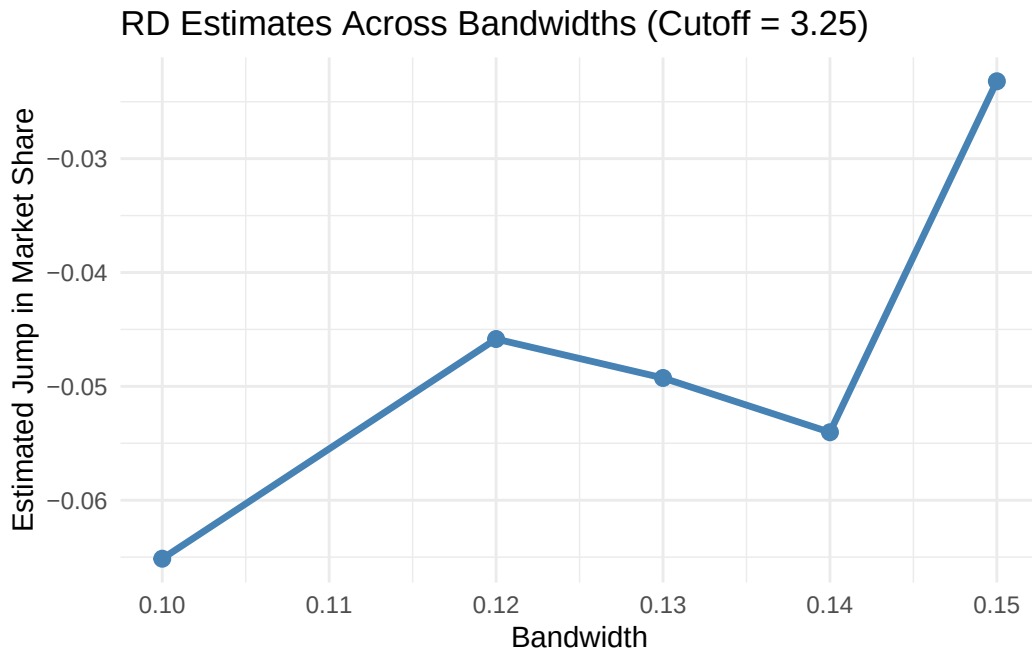
Problem 7

```
# A tibble: 5 x 2
  Bandwidth Estimate
  <dbl>      <dbl>
1    0.1    0.0478
2    0.12   0.0445
3    0.13   0.0482
4    0.14   0.0458
5    0.15   0.0438
```

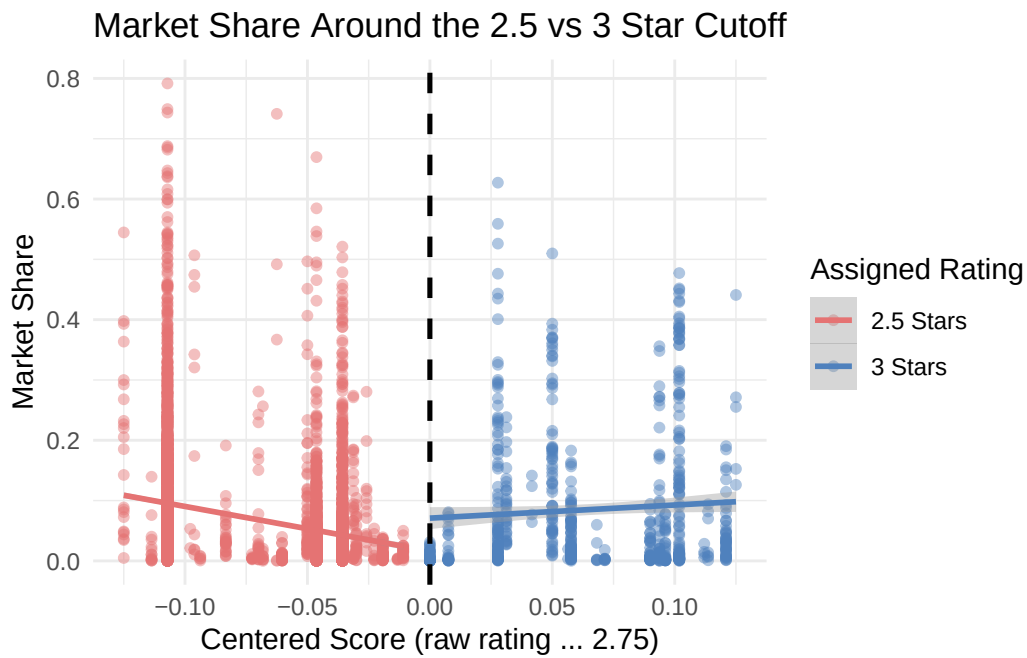
RD Estimates Across Bandwidths (Cutoff = 2.75)



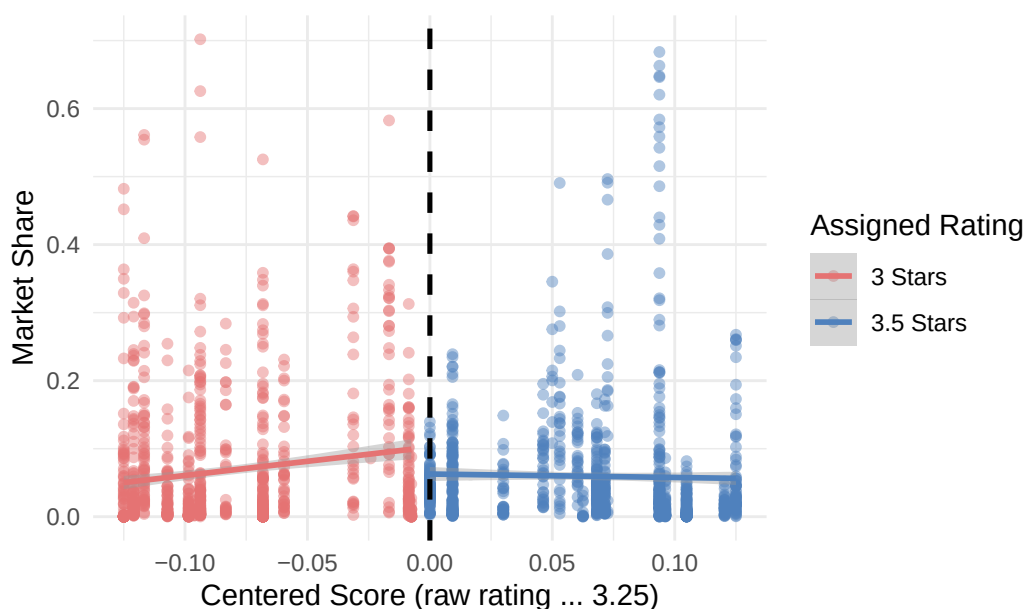
```
# A tibble: 5 x 2
  Bandwidth Estimate
    <dbl>     <dbl>
1    0.1  -0.0651
2    0.12 -0.0458
3    0.13 -0.0493
4    0.14 -0.0540
5    0.15 -0.0232
```



Problem 8



Market Share Around the 3 vs 3.5 Star Cutoff



Problem 9

Table: Plan Characteristics Around the 2.75 Cutoff

```
# A tibble: 2 x 4
  group      share_HMO share_PartD    n
  <chr>      <dbl>      <dbl> <int>
1 Above cutoff  0.447      0.840 2528
2 Below cutoff  0.0194     0.549 22385
```

Table: Plan Characteristics Around the 3.25 Cutoff (3-3.5 Stars)

```
# A tibble: 2 x 4
  group      share_HMO share_PartD    n
  <chr>      <dbl>      <dbl> <int>
1 Above cutoff  0.377      0.817 2452
2 Below cutoff  0.606      0.864 3969
```

These tables show you whether plans just above each cutoff look similar to plans just below it. Plans just above the 2.75 threshold are completely different types of plans as the ones above are much more likely to be HMOs and to offer Part D. Plans for just above the 3.25 threshold are slightly different but to a lesser degree.

Problem 10

Overall, a higher star rating shows more enrollment but the comparison is compromised as plans above and below the threshold are different. Raw ratings tend to be more distributed around the thresholds (2.75 and 3.25), and the adjusted ratings create jumps in star assignments. Because plan characteristics are distinct and not comparable right above or below the cutoff, the RD model cannot cleanly isolate the effect of the star rating alone.